CS 301 · MEETING 12 · A DEEPER LOOK, 3

Hardware Built for the Product

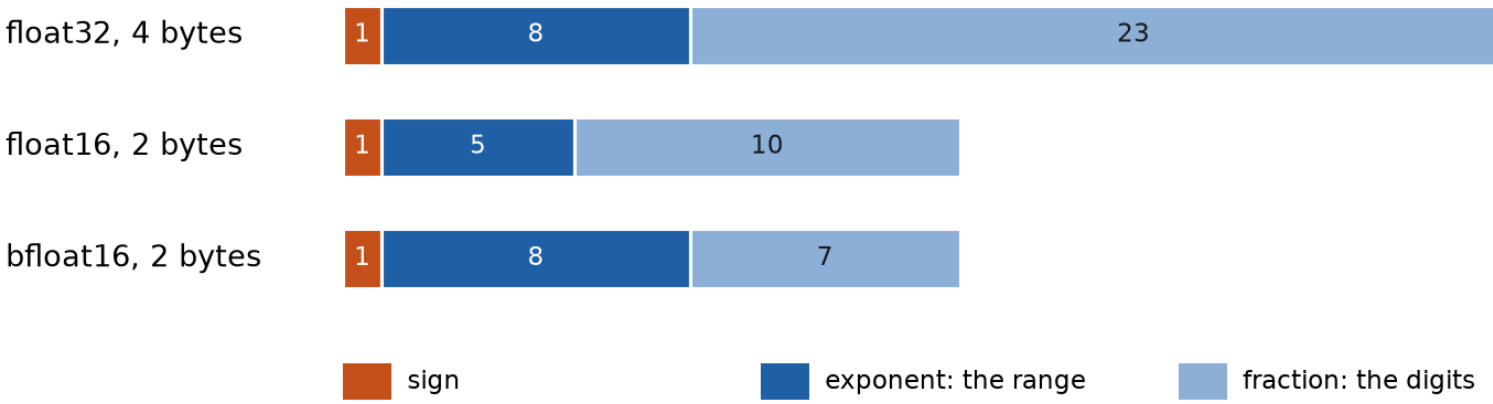
A deeper look: chips designed around X @ W.T, the links that join thousands of them, and software and hardware designed for each other.

Your T4 had a second engine

the T4 multiplies	with	operations a second
32-bit floats: what you measured	2,560 CUDA cores	8.1 trillion
16-bit floats, with 32-bit sums	320 tensor cores	65 trillion
8-bit integers	320 tensor cores	130 trillion
4-bit integers	320 tensor cores	260 trillion

- A **tensor core**: a unit that multiplies small matrices in one step
- Your float32 product never used them

Fewer bits, less energy

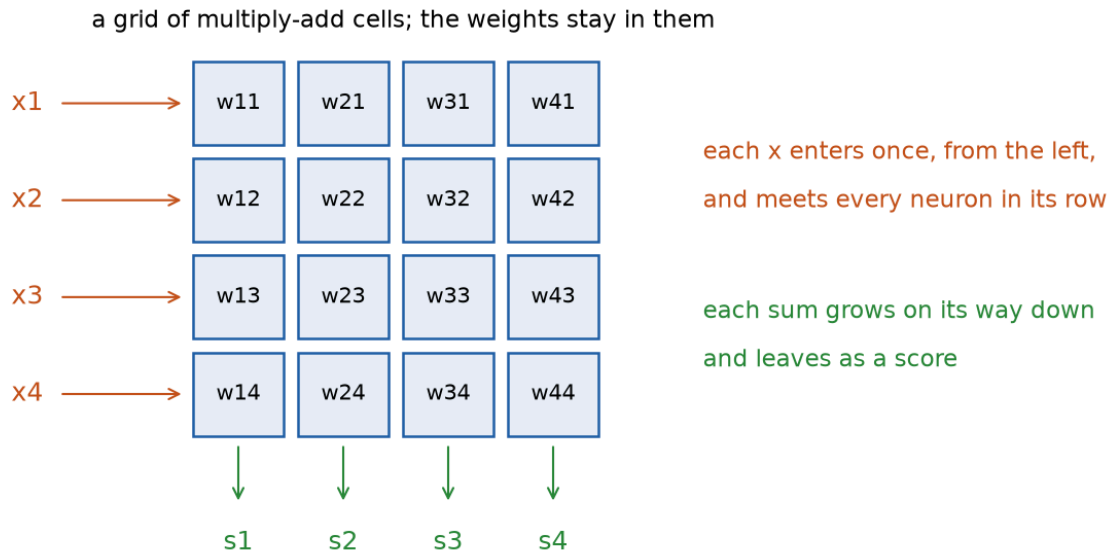


- Half the bytes: twice the numbers for each byte moved
- A multiplication: 3.7 pJ in float32, 1.1 pJ in float16 (2014, 45 nm)
- **bfloat16**: the range of float32 in 2 bytes; made for Google's TPUs

A chip made for the product: the TPU

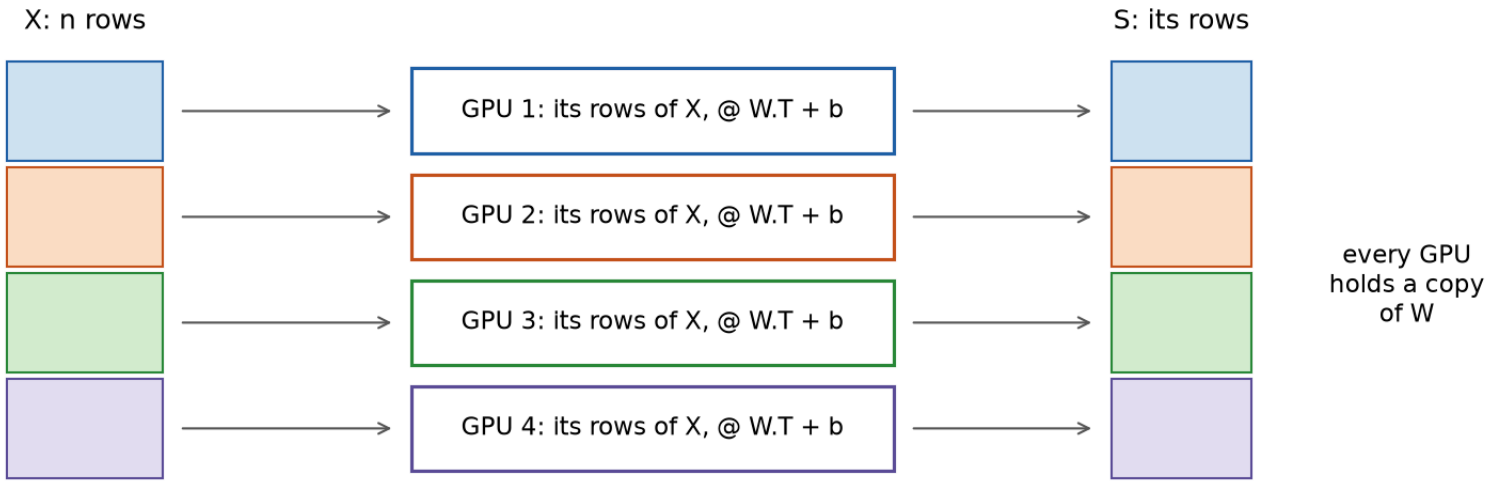
- Google's Tensor Processing Unit, described in 2017
- 65,536 multiply-add units in one square grid, for 8-bit numbers
- 92 trillion operations a second; 28 MiB of memory on the chip
- 15 to 30 times as fast as the GPU and CPU of its day; 30 to 80 times the operations per watt

A systolic array



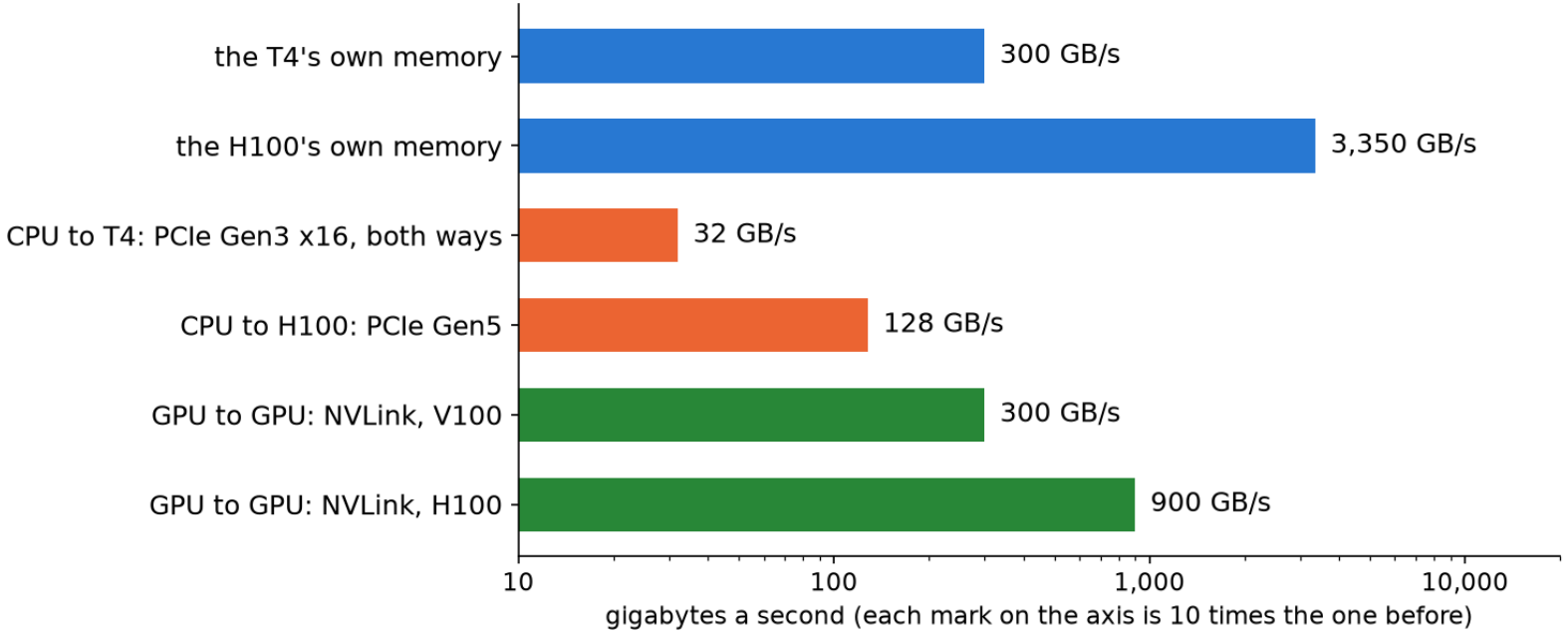
- The weights stay in the cells; the numbers of X flow through
- Each number read from memory once, used by a whole row of cells
- Partial sums pass down the columns between adjacent cells

One GPU is not enough



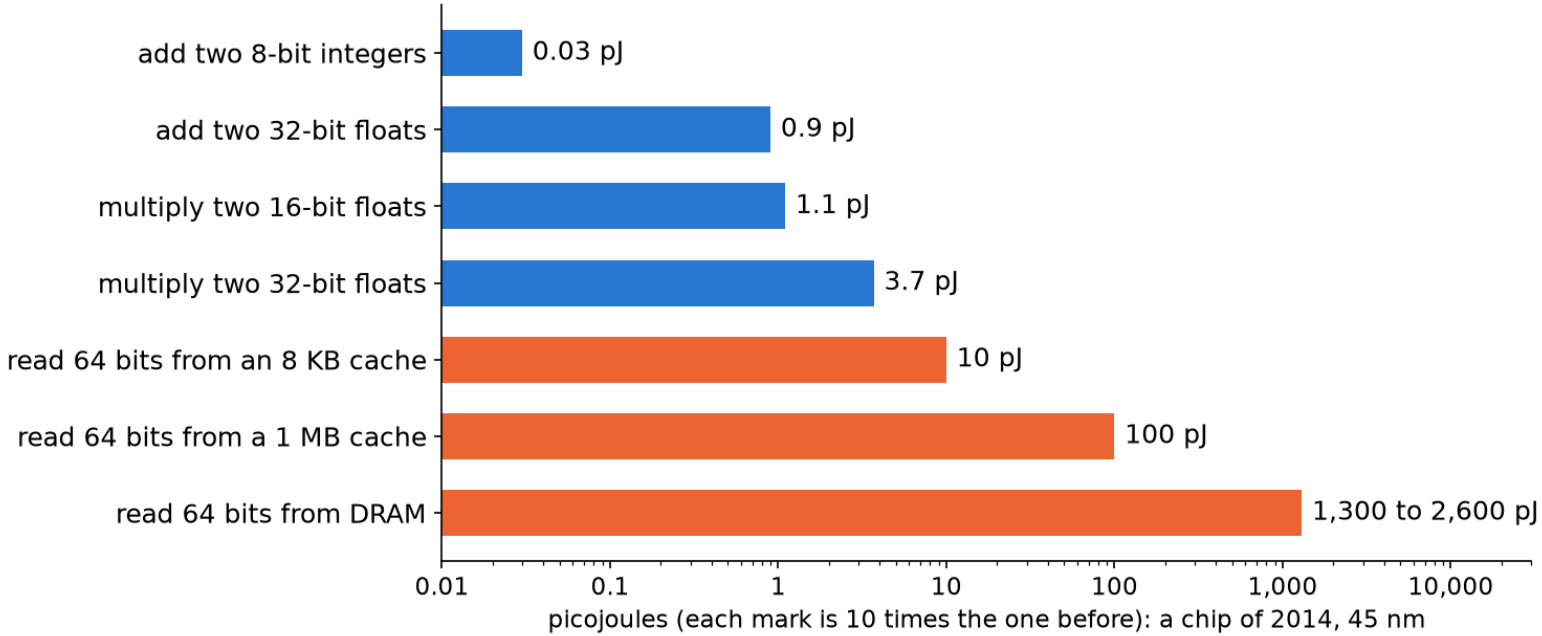
- GPT-3: 3.14×10^{23} operations, on 10,000 GPUs
- Split the rows of X; each GPU computes its rows of S
- Training: the GPUs must exchange numbers at every step

The links between processors



- Local memory and interconnects have different bandwidth limits
- NVLink: a link built only to join GPUs; up to 256 H100s in one system

Moving a number costs more than computing with it



- A read from DRAM: hundreds of multiplications' worth of energy
- Specialized chips keep the numbers next to the multipliers

Software written for the chip

- `X @ W.T` in NumPy calls a **BLAS** library; on a GPU, PyTorch calls cuBLAS
- Rewritten for each processor: its blocks fit that chip's caches and registers
- The same $O(n^3)$ loops, tens of thousands of times faster than three loops in Python

An algorithm written for the memory: FlashAttention

$$\text{softmax}\left(QK^T \right) V$$

- **Attention**, in language models: every word against every word
- FlashAttention (2022): computed block by block in the GPU's on-chip memory
- Same asymptotic arithmetic complexity; less memory traffic

Designed for each other

- **bfloat16**: a number format designed with the chip that uses it
- **2:4 sparsity**: two of every four weights set to zero
- Sparse tensor cores skip the zeros: twice the arithmetic a second
- The H100's data sheet: 1,979 trillion 16-bit operations a second with sparsity, half without

A whole wafer as one chip

- A chip is a small rectangle cut from a disc of silicon, a wafer
- Cerebras WSE-3: the whole wafer as one chip, 46,225 mm^2
- 4 trillion transistors, 900,000 cores
- Its memory on the chip itself, beside the cores

What to take away

- The product of matrices is the workload; chips are now built around it
- Data movement can dominate running time and energy
- Algorithms, number formats and chips are designed together